

Lifted Multiple Shooting for Parallel-in-Time Contact-Rich Trajectory Optimization

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Abstract—Planning through contact requires reasoning about the full dynamics of a robot and its environment, including inertial effects, contact impulses, and friction. While trajectory optimization provides a natural framework for this problem, contact introduces an inner contact-resolution problem whose solution determines the system dynamics and makes gradient-based optimization challenging. In this work, we propose *lifted multiple shooting*, a trajectory optimization formulation that enables planning over fully dynamic contact interactions while retaining the underlying contact-resolution structure. Contact impulses are explicitly represented during optimization, allowing the planner to optimize the complete dynamic trajectory while the contact response remains governed by a separate contact-resolution problem. Our formulation accommodates a broad class of contact constitutive relations and computes their sensitivities through implicit differentiation. To exploit these gradients for efficient planning, we design IP-PAS, a temporally parallelized trajectory optimization algorithm using parallel associative scans that leverages the time-separable structure enabled by our transcription, speeding up planning times on high-dimensional examples. We evaluate it on fully dynamic contact planning tasks, including contact-rich non-prehensile pushing and in-hand manipulation. Across six contact-rich tasks, LMS improves IPOPT success rates by up to 81.5 percentage points over standard multiple shooting, while IP-PAS preserves comparable solution quality to serial methods with up to a $3.1\times$ speedup and scales to horizons of 40,000.

I. INTRODUCTION

Contact-rich manipulation requires robots to plan motion while interacting with their environment. Tasks such as pushing, grasping, and in-hand manipulation couple robot dynamics, contact forces, and friction, making them fundamentally different from collision-free motion planning. Separately planning geometric motion and then accounting for contact can be problematic because contact forces alter the robot motion and resulting interactions. Planning directly over the full dynamics instead optimizes robot motion and contact jointly. Trajectory optimization provides a natural framework, but contact introduces an inner contact-resolution problem that determines the system response. The resulting problem is therefore bilevel: existing approaches either keep contact implicit within shooting methods or incorporate contact conditions directly into the outer optimization, often yielding large complementarity-constrained problems.

We propose *lifted multiple shooting* (LMS), a formulation for trajectory optimization over fully dynamic contact interactions. Contact impulses are explicitly represented during optimization while the underlying contact-resolution problem remains separate. This gives the optimizer direct access to the quantities that determine changes in system momentum while retaining the physical contact model, allowing robot motion

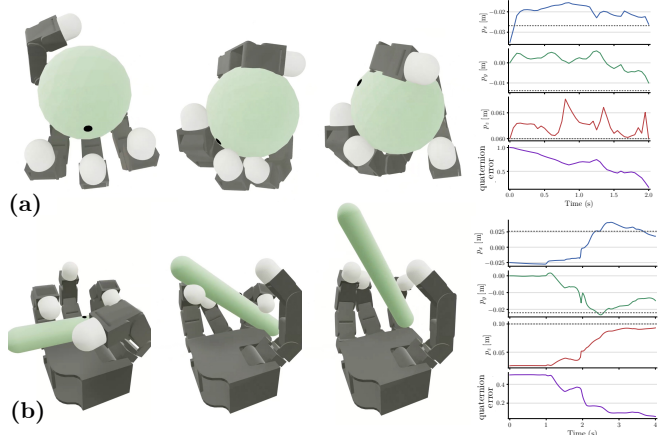


Fig. 1: **In-hand tasks.** We show our solver, IP-PAS, with lifted multiple shooting generating a trajectory for (a) an Allegro hand rotating a sphere 180° over a 40-timestep planning horizon and (b) grasping and lifting a pen from its palm and rotating it such that it rests vertically against the fingers over a 200-timestep horizon. The rightmost column shows absolute 3D position over time (black dotted lines indicate the goal positions) and quaternion goal error.

and contact interactions to be planned simultaneously under the full dynamics. We further introduce a contact solver that accommodates a broad class of contact constitutive relations and computes its sensitivities through implicit differentiation, providing gradients critical for trajectory optimization.

Moreover, the lifted formulation preserves a stagewise optimal-control structure with separable costs and contact constraints and only neighboring-state coupling through the dynamics. We exploit this structure to develop IP-PAS, a GPU-parallel solver based on FATROP [1] that replaces its serial generalized Riccati recursion with an algebraically equivalent parallel associative scan. This parallelizes the contact solves, dynamics derivatives, and Newton-system solve across the horizon, reducing linear-solve depth from $\mathcal{O}(T)$ to $\mathcal{O}(\log T)$. We evaluate our approach on fully dynamic pushing and in-hand manipulation tasks. LMS and IP-PAS address complementary bottlenecks: lifting contact impulses improves optimization reliability, while the preserved stage-wise structure enables temporally parallel optimization for improved computational scalability. Our contributions are:

- We propose *lifted multiple shooting* (LMS) for trajectory optimization over fully dynamic contact interactions, enabling joint planning of motion and contact dynamics.
- We introduce a contact solver that accommodates a broad class of contact constitutive relations and enables efficient gradient-based optimization through contact.
- We develop a novel temporally parallelized trajectory op-

timization algorithm for contact, IP-PAS, that exploits the structure of the proposed transcription.

- We demonstrate fully dynamic contact planning across pushing and in-hand manipulation tasks.

II. RELATED WORK AND PRELIMINARIES

Multibody dynamics with generalized coordinates q and generalized velocity v can be written in discrete time as

$$M(q)(v^+ - v) + \delta t c(q, v) = \delta t \tau(q^+, v^+, u) + J(q)^\top \gamma, \quad (1a)$$

$$q^+ = q \oplus \delta t v^+, \quad (1b)$$

where δt is the timestep, $x = (q, v)$ and $x^+ = (q^+, v^+)$ denote the current and next states, respectively. Here, $M(q)$ is the mass matrix, $c(q, v)$ collects Coriolis, centrifugal, and gravitational forces, and $\tau(q^+, v^+, u)$ represents stiffness, damping, and actuation forces. The matrix $J(q)$ is the contact Jacobian, and γ denotes the contact impulse. The operator $\oplus$ denotes the appropriate integration operation for both vector and quaternion coordinates. Computing the contact Jacobian $J(q)$ requires a closest-point query/collision detection between geometries, which also provides a signed distance $\phi(q)$. When τ is linearized with respect to v , the velocity update can be written explicitly as

$$v^+ = v^* + M_e^{-1} J(q)^\top \gamma, \quad (2)$$

where

$$\begin{aligned} M_e &:= M(q) - \delta t \left(\frac{\partial \tau}{\partial v}(q, v, u) + \delta t \frac{\partial \tau}{\partial q}(q, v, u) \right), \\ v^* &:= v + \delta t M_e^{-1} (\tau(q, v, u) - c(q, v)). \end{aligned} \quad (3)$$

Together, (2) and (1b) define the mapping

$$x^+ = f_{\text{step2}}(x, u, \gamma). \quad (4)$$

The contact impulse γ is not known a priori. Instead, it is determined by solving a contact-resolution problem whose parameters depend on the current state and control (x, u) . Thus, a numerical contact solver induces a mapping

$$\gamma = f_{\text{step1}}(x, u), \quad (5)$$

and the overall dynamics can be viewed as the composition

$$x^+ = f(x, u) := f_{\text{step2}}(x, u, f_{\text{step1}}(x, u)). \quad (6)$$

A. Contact Models

We next review contact models commonly used in planning through contact.

1) *Signorini condition, Coulomb friction, and Maximum dissipation principle*: Under the rigid contact assumption, where rigid bodies cannot interpenetrate, the contact-resolution problem can be written as

$$\begin{aligned} \text{find } & \gamma \\ \text{s.t. } & M_e(v^+ - v^*) = J(q)^\top \gamma, \\ & v_c := \begin{bmatrix} v_n \\ v_{\text{tan}} \end{bmatrix} = J(q) v^+, \\ & \begin{cases} \gamma = 0 & \text{if } v_n + \frac{\phi(q)}{\delta t} > 0, \\ \gamma \in \mathcal{F}_\mu & \text{if } v_c + \begin{bmatrix} \frac{\phi(q)}{\delta t} & 0 & 0 \end{bmatrix}^\top = 0, \\ \gamma_{\text{tan}} = -\mu \gamma_n \frac{v_{\text{tan}}}{\|v_{\text{tan}}\|} & \text{otherwise,} \end{cases} \end{aligned} \quad (7)$$

where $\mathcal{F}_\mu$ is the friction cone with friction coefficient μ .

These conditions define a non-convex, potentially set-valued contact-resolution problem [2]. Thus, the physical model does not uniquely determine γ as a function of (x, u) . A numerical solver selects one solution from this set, thereby inducing the mapping f_{step1} ; neighboring (x, u) values can therefore yield drastically different solver-selected impulses and unreliable gradients.

2) *Log-barrier*: By employing Anitescu's relaxation [3], the contact-resolution problem can be written as

$$\text{find } \gamma \quad (8a)$$

$$\text{s.t. } M_e(v^+ - v^*) = J(q)^\top \gamma, \quad (8b)$$

$$\mathcal{F}_\mu \ni \gamma \perp J(q) v^+ + \begin{bmatrix} \frac{\phi(q)}{\delta t} & 0 & 0 \end{bmatrix}^\top \in \mathcal{F}_\mu^*, \quad (8c)$$

where $\mathcal{F}_\mu^*$ denotes the dual cone of $\mathcal{F}_\mu$. The stationarity condition (8b) and complementarity condition (8c) are precisely the KKT conditions of a convex optimization problem with a log-barrier [4]. However, this contact model still assumes rigid contact and has the hydroplaning artifact, where tangential motion generates a normal force [5].

3) *Complementarity-free*: Another contact model [6] computes the contact impulse in closed form as

$$\gamma = \delta t k \cdot \text{relu}(-\phi(q \oplus \delta t v^*)), \quad (9)$$

where k is the contact stiffness. In other words, the model treats contact as compliant by evaluating the gap distance at a configuration predicted from v^* and setting the contact impulse proportional to the resulting penetration distance.

Because γ is computed independently of v^+ and q^+ , rather than being coupled to the discrete dynamics through an implicit constitutive relation, the method corresponds to an explicit integration scheme. Consequently, it is limited to small contact stiffnesses to maintain numerical stability.

B. Transcriptions for Bilevel Trajectory Optimization

We next review common transcriptions for solving trajectory optimization problems with the dynamics in (6).

1) *Single shooting*: Single shooting transcribes the trajectory optimization problem as

$$\min_{u_{0:T-1}} \sum_{t=0}^{T-1} \ell(x_t, u_t) + \ell_T(x_T), \quad (10)$$

where the dynamics $x_{t+1} = f(x_t, u_t)$ are satisfied by construction by rolling out the system from an initial condition $\hat{x}_0$ under the control sequence $u_{0:T-1}$. Thus, only the controls are optimization variables. Single shooting problems are often solved via differential dynamics programming (DDP).

2) *Multiple shooting*: Multiple shooting instead transcribes the trajectory optimization problem as

$$\begin{aligned} \min_{x_{0:T}, u_{0:T-1}} & \sum_{t=0}^{T-1} \ell(x_t, u_t) + \ell_T(x_T) \\ \text{s.t. } & x_0 = \hat{x}_0, \\ & x_{t+1} = f(x_t, u_t), \quad t = 0, \dots, T-1. \end{aligned} \quad (11)$$

Unlike single shooting, multiple shooting treats the states as optimization variables and enforces the dynamics as equality

TABLE I: Trajectory optimization through contact methods.

Contact model	Single shooting	Multiple shooting	Contact implicit
Signorini condition & Coulomb friction	[9], [10]	[11]	[12], [13], [14], [15], [8], [16]
Log-barrier	[5], [17], [18], [19]	-	[13], [6]
Complementarity-free	-	[6], [20]	-

constraints, allowing dynamically infeasible initial trajectories and often improving conditioning. Contact trajectory optimization can also be viewed as bilevel, with an outer trajectory optimization and an inner contact-resolution problem. In both single and multiple shooting, the contact solver must provide the required sensitivities $\partial f/\partial x$ and $\partial f/\partial u$ to the outer optimization problem so that it can compute a descent direction. In bilevel optimization, such methods are called *descent methods* [7].

3) *Contact implicit*: Contact-implicit trajectory optimization incorporates the KKT conditions of the inner contact-resolution problem directly into the outer trajectory optimization problem. For example, using the log-barrier contact model, the resulting problem takes the form

$$\begin{aligned}
& \min_{\substack{x_0:T, \\ u_0:T-1, \\ \gamma_0:T-1}} \sum_{t=0}^{T-1} \ell(x_t, u_t) + \ell_T(x_T) \\
& \text{s.t. } x_0 = \hat{x}_0, \\
& x_{t+1} = f_{\text{step2}}(x_t, u_t, \gamma_t), \quad t = 0, \dots, T-1, \\
& \mathcal{F}_\mu \ni \gamma_t \perp J(q_t) v_{t+1} + [\frac{\phi(q_t)}{\delta t} \quad 0 \quad 0]^\top \in \mathcal{F}_\mu^*, \\
& t = 0, \dots, T-1.
\end{aligned} \tag{12}$$

Rather than solving the contact-resolution problem separately at each timestep, this formulation incorporates the contact impulses and feasibility conditions directly into the trajectory optimization, resulting in a large mathematical program with equilibrium constraints (MPEC). In bilevel optimization, this is a *KKT single-level reduction* [7].

Beyond the difficulty of solving an MPEC, the problem can grow substantially when $J(q)$ and $\phi(q)$ lack closed-form expressions for geometries such as boxes. In these cases, contact forces are introduced for each candidate face, together with complementarity constraints enforcing that at most one face is active [8]. In summary, existing approaches to trajectory optimization through contact, whether termed *contact smoothing* or *contact implicit*, solve the same underlying bilevel trajectory optimization problem using different mathematical transcriptions. Table I categorizes prior work.

This raises the question of whether a better transcription can preserve the bilevel structure without reducing it to a single-level problem, while improving optimization reliability to avoid poor local minima from naive warm starts. In the following, we will outline our novel transcription, *lifted multiple shooting* (LMS) (Sec. III), and our parallel-in-time solver for efficiently solving LMS trajectory optimization problems that leverages the multiple-shooting structure of LMS (Sec. IV).

III. PROBLEM TRANSCRIPTION AND CONTACT SOLVER

We transcribe the trajectory optimization problem as

$$\min_{\substack{x_0:T, u_0:T-1, \\ \gamma_0:T-1}} \sum_{t=0}^{T-1} \ell(x_t, u_t) + \ell_T(x_T) \tag{13a}$$

$$\text{s.t. } x_0 = \hat{x}_0, \tag{13b}$$

$$x_{t+1} = f_{\text{step2}}(x_t, u_t, \gamma_t), \quad t = 0, \dots, T-1, \tag{13c}$$

$$\gamma_t = f_{\text{step1}}(x_t, u_t), \quad t = 0, \dots, T-1, \tag{13d}$$

$$\gamma_t \in \mathcal{F}_\mu, \quad t = 0, \dots, T-1, \tag{13e}$$

$$g(x_t, u_t) \leq 0, \quad t = 0, \dots, T-1. \tag{13f}$$

We call this transcription *lifted multiple shooting*. In contrast to multiple shooting, contact impulses are lifted out as explicit decision variables, with (13e) enforcing that they lie within the friction cone. In contrast to contact implicit, the bilevel structure is preserved: the contact-resolution problem defining $f_{\text{step1}}(x_t, u_t)$ is solved separately at each timestep rather than being lumped into a single outer optimization problem. Inequality (13f) represents additional stage constraints.

We next describe how $\gamma = f_{\text{step1}}(x, u)$ is implicitly defined. At each timestep, the contact impulse is obtained by solving a convex contact-resolution problem. The contact-resolution problem is

$$\min_{v^+} \frac{1}{2}(v^+ - v^*)^\top M_e(v^+ - v^*) + \ell_c(v_c), \tag{14}$$

where M_e and v^* are defined in (3), $v_c := J(q) v^+$ is the contact velocity and ℓ_c is a convex regularization term that will be defined later.

The first-order optimality condition of (14) is

$$M_e(v^+ - v^*) + J(q)^\top \frac{\partial \ell_c(v_c)}{\partial v_c} = 0. \tag{15}$$

Comparing (15) with the discrete-time dynamics in (2), we obtain the relationship between the contact impulse and the regularization term ℓ_c :

$$\gamma = -\frac{\partial \ell_c(v_c)}{\partial v_c}. \tag{16}$$

Thus, for a constitutive relation $\gamma = \gamma(v_c)$, ℓ_c is defined by $\nabla_{v_c} \ell_c(v_c) = -\gamma(v_c)$. Since M_e is positive definite and ℓ_c is convex whenever $-\partial \gamma(v_c)/\partial v_c \succeq 0$, the objective in (14) is strongly convex in v^+ . Therefore, the contact-resolution problem admits a unique optimal velocity v^+ . The corresponding contact impulse is obtained from (16), which implicitly defines the mapping $\gamma = f_{\text{step1}}(x, u)$.

Although the contact-resolution problem is formulated in terms of the convex regularization term ℓ_c , its explicit evaluation is not required by our solver. We use Newton's method to solve the first-order optimality condition

$$r(v^+; x, u) := M_e(v^+ - v^*) - J(q)^\top \gamma(v_c) = 0, \tag{17}$$

A Newton direction is obtained by solving

$$H \Delta v^+ = -r, \quad H := M_e - J(q)^\top \frac{\partial \gamma}{\partial v_c} J(q). \tag{18}$$

An exact line search determines the step size by finding the zero of the directional derivative, which can likewise be evaluated using only the constitutive relation $\gamma(v_c)$ and

its derivative [21]. Thus, explicit evaluation of the scalar potential ℓ_c is not required by the solver.

The same Hessian enables efficient implicit differentiation of the contact solution. Since the optimal velocity satisfies $r(v^+; x, u) = 0$, differentiating this equation with respect to any parameter $p \in \{x, u\}$ gives

$$H \frac{\partial v^+}{\partial p} = -\frac{\partial r}{\partial p}. \quad (19)$$

The right-hand side is obtained by forward-mode automatic differentiation of the residual function, while the sensitivities are computed by solving a linear system using the already factorized Hessian H . Thus, implicit differentiation avoids differentiating through the Newton iterations and makes computing the contact sensitivities inexpensive relative to solving the contact problem itself.

For the contact constitutive relation, we use the Hunt-Crossley model [22] for the normal contact impulse. For a contact with current signed distance $\phi_0 := \phi(q)$ and relative normal velocity v_n , the impulse is given by

$$\gamma_n = \delta t k \cdot \text{relu}_\epsilon(-\phi_0 - \delta t v_n) \cdot \text{relu}(1 - dv_n), \quad (20)$$

where k and d are the contact stiffness and Hunt-Crossley damping, respectively. Here, relu_ϵ denotes a smooth approximation of the rectified linear unit. We use

$$\text{relu}_\epsilon(-\phi) = \begin{cases} \frac{\epsilon^2}{4(\epsilon + \phi)}, & \phi \geq -\epsilon/2 \\ -\phi, & \phi < -\epsilon/2. \end{cases} \quad (21)$$

Smoothing is introduced as a necessary numerical compromise to enable gradient-based optimization through contact dynamics. In the lifted multiple shooting transcription (13), the equality constraint $\gamma_t = f_{\text{step1}}(x_t, u_t)$ must be corrected whenever it is violated. For a nonsmooth contact model, however, f_{step1} may have zero gradient, providing insufficient first-order information for the optimizer to reduce the constraint violation. We observe that the lifted multiple shooting transcription requires substantially less smoothing than single- or multiple-shooting transcriptions: in our experiments, it operates reliably with a smoothing parameter approximately $1000\times$ smaller. This substantially reduces the deviation of the smoothed contact model from the original nonsmooth dynamics.

IV. PARALLEL-IN-TIME SOLVER

We introduce an IP-PAS (interior-point parallel associative scan) solver for solving (13). The solver algorithm is based on FATROP [1] but utilizes temporal parallelization. Specifically, the temporal recursions admit an associative formulation with $\mathcal{O}(\log T)$ computational depth and $\mathcal{O}(T)$ total work, yielding a parallel runtime of $\mathcal{O}(T/N_{\text{core}} + \log T)$ on N_{core} compute cores, enabling long-horizon planning.

A. Stagewise Optimal-Control Structure

Define the augmented stage decision $w_t \triangleq [u_t^\top \ \gamma_t^\top]^\top$. Then (13) has the generic stagewise form

$$\min_{x_0:T, w_0:T-1} \sum_{t=0}^{T-1} \ell_t(x_t, w_t) + \ell_T(x_T) \quad (22a)$$

$$\text{s.t. } x_0 = \hat{x}_0, \quad (22b)$$

$$x_{t+1} = f_t(x_t, w_t), \quad (22c)$$

$$e_t(x_t, w_t) = 0, \quad (22d)$$

$$\underline{h}_t \leq h_t(x_t, w_t) \leq \bar{h}_t, \quad (22e)$$

where $f_t(x_t, w_t) = f_{\text{step2}}(x_t, u_t, \gamma_t)$ and $e_t(x_t, w_t) = \gamma_t - f_{\text{step1}}(x_t, u_t)$. The two-sided inequality map h_t collects actuator and contact-impulse bounds, friction-cone inequalities, and any additional stage constraints. For example, for an elliptic Coulomb cone, our implementation includes the smooth row $\|\text{diag}(\mu_t)^{-1} \gamma_{t,\text{tan}}\|_2^2 - \gamma_{t,n}^2 \leq 0$, in addition to $\gamma_{t,n} \geq 0$. Here μ_t denotes the vector of tangential friction coefficients at stage t . The cost, dynamics, equalities, and inequalities in (22) are *separable by stage* except for the nearest-neighbor dynamics constraints. This is precisely the optimal-control structure exploited by FATROP: its primal-dual interior-point method retains the numerical robustness of a filter line-search method while efficiently solving each Newton system using a generalized Riccati recursion that can exploit the sparsity of the cost and constraints of (22). This structure is especially favorable here because the contact solve f_{step1} , the state update f_{step2} , and their derivatives are already evaluated in JAX on the GPU. We therefore develop a JAX implementation of the FATROP Newton system whose temporal linear algebra also remains on the GPU.

B. Parallel Associative Scan for FATROP

Despite its efficiency, FATROP's generalized Riccati recursion is *linear* in the horizon and proceeds *serially* across time [1]. Our main solver contribution is to derive an algebraically equivalent parallel associative scan (PAS) for FATROP's reduced primal-dual Newton system. PAS has previously enabled logarithmic-depth temporal parallelization for LQR and related trajectory-optimization problems [23], [24], overcoming their originally serial nature. Here, we extend this PAS construction to the stagewise-separable equality and two-sided inequality structure used by FATROP.

1) *Reduced primal-dual Newton system:* At a primal-dual interior-point iterate, we linearize the dynamics (22c), stage equalities (22d), and inequalities (22e) as

$$\Delta x_{t+1} = A_t \Delta x_t + B_t \Delta w_t + a_t, \quad (23)$$

$$0 = E_t \Delta x_t + F_t \Delta w_t + e_t, \quad (24)$$

$$\Delta h_t = G_t^x \Delta x_t + G_t^w \Delta w_t. \quad (25)$$

Define the stage Lagrangian

$$\begin{aligned} \mathcal{L}_t(x_t, w_t) &\triangleq \ell_t(x_t, w_t) + \lambda_{t+1}^\top f_t(x_t, w_t) \\ &\quad + \beta_t^\top e_t(x_t, w_t) + \nu_t^\top h_t(x_t, w_t), \end{aligned} \quad (26)$$

where λ_{t+1} , β_t , and ν_t are the multipliers associated with the dynamics, stage equalities, and inequality constraints,

respectively. Here λ_0 denotes the multiplier associated with the initial-condition constraint $x_0 = \hat{x}_0$. We partition the Hessian of $\mathcal{L}_t$ as

$$\nabla_{(x_t, w_t)}^2 \mathcal{L}_t = \begin{bmatrix} Q_t & S_t \\ S_t^\top & R_t \end{bmatrix}, \text{ with} \quad (27)$$

$$Q_t = \nabla_{x_t x_t}^2 \mathcal{L}_t, \quad S_t = \nabla_{x_t w_t}^2 \mathcal{L}_t, \quad R_t = \nabla_{w_t w_t}^2 \mathcal{L}_t. \quad (28)$$

The state and stage-variable stationarity residuals are

$$q_t \triangleq \nabla_{x_t} \ell_t + A_t^\top \lambda_{t+1} + E_t^\top \beta_t + (G_t^x)^\top \nu_t - \lambda_t, \quad (29)$$

$$r_t \triangleq \nabla_{w_t} \ell_t + B_t^\top \lambda_{t+1} + F_t^\top \beta_t + (G_t^w)^\top \nu_t. \quad (30)$$

FATROP introduces an inequality slack s_t together with lower- and upper-bound dual variables z_t^L, z_t^U . For the finite lower and upper bounds, define the positive margins

$$D_t^L = s_t - \underline{h}_t, \quad D_t^U = \bar{h}_t - s_t. \quad (31)$$

Eliminating the slack and bound-dual Newton directions gives the diagonal coefficient

$$\Sigma_t \triangleq \text{diag} \left(\frac{z_t^L}{D_t^L} + \frac{z_t^U}{D_t^U} \right), \quad W_t \triangleq \Sigma_t^{-1}, \quad (32)$$

where the divisions in (32) are elementwise and absent one-sided bounds contribute zero. FATROP also uses inertia regularization $\delta_w \geq 0$, which enters the primal Hessian blocks Q_t^δ and R_t^δ . The remaining local primal-dual equations can be written as

$$M_t = \begin{bmatrix} R_t^\delta & F_t^\top & (G_t^w)^\top \\ F_t & 0 & 0 \\ G_t^w & 0 & -W_t \end{bmatrix}, \quad z_t = \begin{bmatrix} \Delta w_t \\ \Delta \beta_t \\ \Delta \nu_t \end{bmatrix}, \quad (33)$$

$$K_t = \begin{bmatrix} S_t^\top \\ E_t \\ G_t^x \end{bmatrix}, \quad L_t = \begin{bmatrix} B_t^\top \\ 0 \\ 0 \end{bmatrix}, \quad d_t = \begin{bmatrix} r_t \\ e_t \\ \tilde{g}_t \end{bmatrix}.$$

with the local primal-dual equation therefore written as

$$M_t z_t + K_t \Delta x_t + L_t \Delta \lambda_{t+1} + d_t = 0. \quad (34)$$

Here $R_t^\delta = R_t + \delta_w I$, $\Delta \beta_t$ is the stage-equality dual direction, $\Delta \nu_t$ is the reduced inequality-dual direction, and $\tilde{g}_t$ collects the primal, dual, and complementarity residuals after slack elimination. The state-stationarity equation is

$$\Delta \lambda_t = Q_t^\delta \Delta x_t + S_t \Delta w_t + A_t^\top \Delta \lambda_{t+1} + E_t^\top \Delta \beta_t + (G_t^x)^\top \Delta \nu_t + q_t, \quad (35)$$

with $Q_t^\delta = Q_t + \delta_w I$.

Because W_t is diagonal, our implementation does not factor the full matrix M_t . From the last row of (34),

$$\Delta \nu_t = \Sigma_t (G_t^w \Delta w_t + G_t^x \Delta x_t + \tilde{g}_t), \quad (36)$$

so the dense stage solve reduces to

$$\begin{bmatrix} R_t^\delta + (G_t^w)^\top \Sigma_t G_t^w & F_t^\top \\ F_t & 0 \end{bmatrix}, \quad (37)$$

of dimension $n_w + n_e$, where $n_w := \dim(w_t)$ and $n_e := \dim(e_t)$, independent of the number of inequality rows. The quantities involving G_t^x and $\tilde{g}_t$ enter only as Schur-complement updates. This compact representation is important for contact-rich problems, where the number of contact-impulse bounds and friction constraints can be large.

2) *Stage elimination and boundary map*: For each stage, FATROP solves (34) conditionally on its two boundary variables, Δx_t and $\Delta \lambda_{t+1}$:

$$z_t = - (Z_t^x \Delta x_t + z_t^0 + Z_t^\lambda \Delta \lambda_{t+1}), \quad (38)$$

where $Z_t^x = M_t^{-1} K_t$, $z_t^0 = M_t^{-1} d_t$, and $Z_t^\lambda = M_t^{-1} L_t$; in implementation these products are obtained through the reduced system (37). Define

$$\bar{B}_t \triangleq [B_t \quad 0 \quad 0]. \quad (39)$$

Substituting (38) into (23) and (35) gives the fixed-size boundary relation

$$\Delta \lambda_t = P_t \Delta x_t + p_t + \bar{A}_t^\top \Delta \lambda_{t+1}, \quad (40a)$$

$$\Delta x_{t+1} = \bar{A}_t \Delta x_t + b_t - C_t \Delta \lambda_{t+1}, \quad (40b)$$

where

$$P_t = Q_t^\delta - K_t^\top Z_t^x, \quad p_t = q_t - K_t^\top z_t^0, \\ C_t = \bar{B}_t Z_t^\lambda, \quad \bar{A}_t = A_t - \bar{B}_t Z_t^x, \quad b_t = a_t - \bar{B}_t z_t^0. \quad (41)$$

Thus, regardless of the number of controls, equalities, or inequalities at a stage, all information needed to connect adjacent stages is summarized by the five state-sized quantities

$$\mathcal{E}_{t,t+1} \triangleq (P_t, p_t, C_t, \bar{A}_t, b_t). \quad (42)$$

This is the key reduction that makes the FATROP Newton system amenable to PAS.

3) *Associative composition*: The boundary relation (40) is equivalently represented by the dual conditional value function

$$V_{i \rightarrow j}(\Delta x_i, \Delta x_j) = \max_{\eta} \left\{ \frac{1}{2} \Delta x_i^\top P_{i,j} \Delta x_i + p_{i,j}^\top \Delta x_i - \frac{1}{2} \eta^\top C_{i,j} \eta - \eta^\top (\Delta x_j - A_{i,j} \Delta x_i - b_{i,j}) \right\}. \quad (43)$$

This generalizes the temporal-parallel LQR representation in [23]. Composing two neighboring segments means eliminating their shared boundary state,

$$V_{i \rightarrow j} = \min_{\Delta x_k} (V_{i \rightarrow k} + V_{k \rightarrow j}). \quad (44)$$

If $\mathcal{E}_{i,k} = (P_1, p_1, C_1, A_1, b_1)$ and $\mathcal{E}_{k,j} = (P_2, p_2, C_2, A_2, b_2)$, direct Schur elimination gives

$$P = P_1 + A_1^\top (I + P_2 C_1)^{-1} P_2 A_1, \quad (45a)$$

$$p = p_1 + A_1^\top (I + P_2 C_1)^{-1} (p_2 + P_2 b_1), \quad (45b)$$

$$A = A_2 (I + C_1 P_2)^{-1} A_1, \quad (45c)$$

$$C = C_2 + A_2 (I + C_1 P_2)^{-1} C_1 A_2^\top, \quad (45d)$$

$$b = b_2 + A_2 (I + C_1 P_2)^{-1} (b_1 - C_1 p_2). \quad (45e)$$

The resulting tuple (P, p, C, A, b) is exactly the boundary map for the combined segment $i \rightarrow j$. Because composition corresponds to eliminating internal boundary variables, the operator is associative [23]. Hence all suffix maps can be computed with a reverse parallel scan in $\mathcal{O}(\log T)$ temporal depth rather than the $\mathcal{O}(T)$ depth of a serial Riccati recursion [25]. We append the terminal element, where

$Q_T^\delta = \nabla_{x_T x_T}^2 \ell_T + \delta_w I$ and $q_T = \nabla_{x_T} \ell_T - \lambda_T$ denote the regularized terminal-cost Hessian and terminal stationarity residual, respectively:

$$\mathcal{E}_T = (Q_T^\delta, q_T, 0, 0, 0). \quad (46)$$

We then apply the scan to $\{\mathcal{E}_{0,1}, \dots, \mathcal{E}_{T-1,T}, \mathcal{E}_T\}$. The resulting suffix element at $t+1$ provides the value Hessian and gradient (P_{t+1}^V, p_{t+1}^V) required by every stage.

4) *Parallel forward recovery*: Combining (40b) with $\Delta \lambda_{t+1} = P_{t+1}^V \Delta x_{t+1} + p_{t+1}^V$ yields an affine state recursion

$$\Delta x_{t+1} = k_t + K_t^x \Delta x_t, \text{ with} \quad (47)$$

$$\begin{aligned} K_t^x &= (I + C_t P_{t+1}^V)^{-1} \bar{A}_t, \\ k_t &= (I + C_t P_{t+1}^V)^{-1} (b_t - C_t p_{t+1}^V). \end{aligned} \quad (48)$$

Affine maps are themselves associative:

$$(k_2, K_2) \circ (k_1, K_1) = (k_2 + K_2 k_1, K_2 K_1). \quad (49)$$

A second, forward PAS therefore recovers all Δx_t in logarithmic temporal depth. We then evaluate all costates $\Delta \lambda_{t+1} = P_{t+1}^V \Delta x_{t+1} + p_{t+1}^V$ and recover $\Delta w_t, \Delta \beta_t, \Delta \nu_t$ from (38) independently across stages. Finally, the original slack and lower/upper bound-dual directions are recovered elementwise from the interior-point complementarity equations. FATROP also increases δ_w until the reduced Hessian has the required inertia. The reverse PAS already provides every downstream value Hessian P_{t+1}^V , so this test remains parallel. Specifically, with N_t spanning the nullspace of F_t , our implementation checks

$$N_t^\top (R_t + \delta_w I + B_t^\top P_{t+1}^V B_t + (G_t^w)^\top \Sigma_t G_t^w) N_t \succ 0 \quad (50)$$

for all stages in parallel, together with full row rank of F_t . Thus, the PAS preserves FATROP's reduced-Hessian regularization criterion without adding a serial sweep.

C. JAX/GPU Implementation

Our implementation retains FATROP's nonlinear primal-dual policy, including slack and multiplier initialization, monotone barrier updates, inertia correction, filter line search, second-order correction, and restoration, while making the reduced Newton solve interchangeable between a serial generalized Riccati recursion and the PAS backend above. Thus, the two backends differ only in *how* the same linearized KKT system is solved (i.e., Riccati or PAS), which provides a direct correctness and runtime comparison.

All stage derivatives are generated with JAX automatic differentiation. On GPU, the contact-resolution map, explicit state update, costs, and constraints are vectorized over the horizon, so their derivatives are evaluated concurrently. The PAS backend then vectorizes the local factorizations (37), executes the reverse boundary-map scan (45), executes the forward affine scan (49), and recovers all remaining Newton variables in parallel. The diagonal matrix W_t is stored as a vector, and each PAS tree node factors $I + P_2 C_1$ once and solves all required right-hand sides together. These choices avoid dense inequality factorizations and repeated stagewise

TABLE II: Benchmark results. Best and second-best values are shown in blue and green, respectively.

Task (Horizon)	Solver	Transcription	Success rate [%]	Goal error	Solve time [s]
Push box (80)	IPOPT	Single shooting	0.0	0.4238 $\pm$ 0.1338	0.94 $\pm$ 0.54
		Multiple shooting	40.0	0.2573 $\pm$ 0.2375	2.30 $\pm$ 0.84
		Contact implicit	47.5	0.2243 $\pm$ 0.2375	1.21 $\pm$ 0.47
	FATROP	Lifted multiple s.	98.5	0.0129 $\pm$ 0.0550	2.43 $\pm$ 0.58
		Multiple shooting	33.5	0.2640 $\pm$ 0.2208	1.47 $\pm$ 0.22
		Lifted multiple s.	96.0	0.0237 $\pm$ 0.0954	1.58 $\pm$ 0.26
Push circle (100)	IPOPT	Single shooting	0.0	0.1733 $\pm$ 0.0449	1.19 $\pm$ 0.44
		Multiple shooting	33.5	0.1334 $\pm$ 0.0698	2.25 $\pm$ 0.45
		Contact implicit	25.5	0.1503 $\pm$ 0.0642	0.29 $\pm$ 0.08
	FATROP	Lifted multiple s.	95.5	0.0212 $\pm$ 0.0608	2.64 $\pm$ 0.29
		Multiple shooting	34.0	0.1295 $\pm$ 0.0690	1.26 $\pm$ 0.18
		Lifted multiple s.	100.0	0.0049 $\pm$ 0.0094	1.68 $\pm$ 0.22
Square knob (400)	IPOPT	Single shooting	0.0	0.7010 $\pm$ 0.8527	3.42 $\pm$ 1.07
		Multiple shooting	61.5	0.2728 $\pm$ 0.4402	3.92 $\pm$ 1.28
		Contact implicit	92.0	0.2652 $\pm$ 1.4263	1.69 $\pm$ 0.21
	FATROP	Lifted multiple s.	100.0	0.0120 $\pm$ 0.0081	3.49 $\pm$ 0.69
		Multiple shooting	67.5	0.2435 $\pm$ 0.4540	1.86 $\pm$ 0.21
		Lifted multiple s.	94.0	0.0921 $\pm$ 0.2730	1.98 $\pm$ 0.51
Planar finger (100)	IPOPT	Single shooting	0.0	0.5814 $\pm$ 0.0487	2.33 $\pm$ 0.62
		Multiple shooting	89.5	0.0551 $\pm$ 0.0981	2.68 $\pm$ 0.79
		Contact implicit	35.0	0.4489 $\pm$ 0.0413	1.08 $\pm$ 0.21
	FATROP	Lifted multiple s.	99.5	0.0120 $\pm$ 0.0136	2.17 $\pm$ 0.45
		Multiple shooting	69.0	0.3683 $\pm$ 0.0854	1.67 $\pm$ 0.21
		Lifted multiple s.	96.5	0.0105 $\pm$ 0.0219	2.00 $\pm$ 0.31
Sphere in-hand (20)	IPOPT	Single shooting	7.0	0.3155 $\pm$ 0.5281	8.06 $\pm$ 2.86
		Multiple shooting	13.0	0.2734 $\pm$ 0.5047	10.5 $\pm$ 3.84
		Contact implicit	14.0	0.5026 $\pm$ 0.4008	7.16 $\pm$ 1.06
	FATROP	Lifted multiple s.	94.5	0.0056 $\pm$ 0.0239	15.3 $\pm$ 3.00
		Multiple shooting	33.0	0.3472 $\pm$ 0.7692	9.02 $\pm$ 5.32
		Lifted multiple s.	90.5	0.0211 $\pm$ 0.0579	12.6 $\pm$ 4.21
Pen in-hand (20)	IPOPT	Single shooting	0.0	0.2303 $\pm$ 0.9982	8.26 $\pm$ 2.66
		Multiple shooting	20.0	0.1647 $\pm$ 0.4306	8.37 $\pm$ 2.92
		Contact implicit	32.5	0.1355 $\pm$ 0.3989	7.61 $\pm$ 1.08
	FATROP	Lifted multiple s.	95.5	0.0437 $\pm$ 0.0526	21.6 $\pm$ 4.06
		Multiple shooting	22.0	0.1013 $\pm$ 0.3099	8.46 $\pm$ 2.97
		Lifted multiple s.	91.0	0.0763 $\pm$ 0.0801	17.6 $\pm$ 1.04
Pen in-hand (20)	IPOPT	Single shooting	0.0	0.2303 $\pm$ 0.9982	8.26 $\pm$ 2.66
		Multiple shooting	20.0	0.1647 $\pm$ 0.4306	8.37 $\pm$ 2.92
		Contact implicit	32.5	0.1355 $\pm$ 0.3989	7.61 $\pm$ 1.08
	FATROP	Lifted multiple s.	95.5	0.0437 $\pm$ 0.0526	21.6 $\pm$ 4.06
		Multiple shooting	22.0	0.1013 $\pm$ 0.3099	8.46 $\pm$ 2.97
		Lifted multiple s.	91.0	0.0763 $\pm$ 0.0801	17.6 $\pm$ 1.04
Pen in-hand (20)	IPOPT	Single shooting	0.0	0.2303 $\pm$ 0.9982	8.26 $\pm$ 2.66
		Multiple shooting	20.0	0.1647 $\pm$ 0.4306	8.37 $\pm$ 2.92
		Contact implicit	32.5	0.1355 $\pm$ 0.3989	7.61 $\pm$ 1.08
	FATROP	Lifted multiple s.	95.5	0.0437 $\pm$ 0.0526	21.6 $\pm$ 4.06
		Multiple shooting	22.0	0.1013 $\pm$ 0.3099	8.46 $\pm$ 2.97
		Lifted multiple s.	91.0	0.0763 $\pm$ 0.0801	17.6 $\pm$ 1.04

synchronization. The resulting architecture is particularly well matched to contact planning: the expensive contact dynamics and their derivatives already execute on the GPU, while the PAS removes the serial-in-time Riccati solve that would otherwise interrupt this GPU-parallel pipeline. With sufficient parallel hardware, the horizon-dependent depth of the linear solve is reduced from $\mathcal{O}(T)$ to $\mathcal{O}(\log T)$ while retaining $\mathcal{O}(T)$ total work.

V. RESULTS AND DISCUSSION

Experimental Setup. We evaluate the proposed lifted multiple shooting transcription and IP-PAS solver on six contact planning tasks: box pushing (Fig. 2a), circle pushing (Fig. 2b), square knob manipulation (Fig. 2c), planar finger manipulation (Fig. 2d), sphere in-hand manipulation (Fig. 1a), and pen in-hand manipulation (Fig. 1b). The in-hand benchmarks in Table II use a 20-step horizon and are

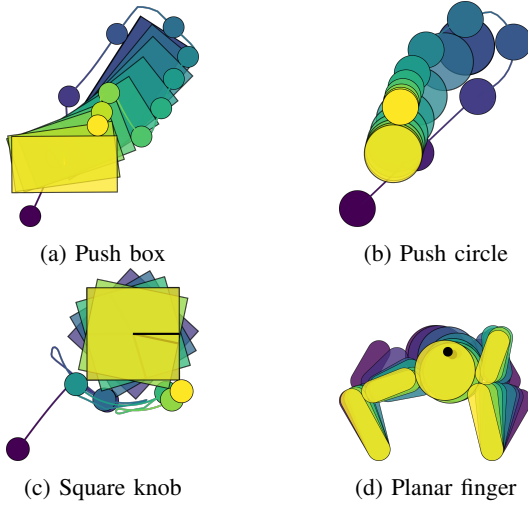


Fig. 2: Planar benchmark tasks.

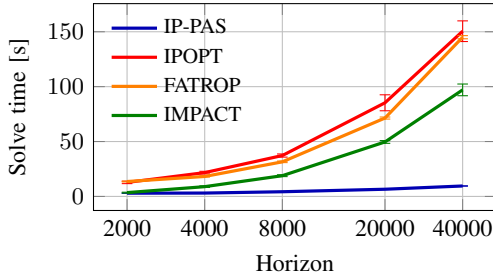


Fig. 3: Solve time scaling with horizon length. IPOPT, FATROP, and IP-PAS use LMS; IMPACT is contact implicit.

evaluated separately from the longer-horizon IP-PAS examples in Fig. 1. For each task and solver, we evaluate 200 randomly sampled initial conditions. All methods are initialized with a zero trajectory and receive no task-specific warm start. Goal error is the terminal cost $\ell_T(x_T)$.

We compare single shooting, multiple shooting, lifted multiple shooting, and contact-implicit formulations using IPOPT [26]. We then compare IP-PAS with serial FATROP [1] under lifted multiple shooting to isolate temporal parallelization, and with IMPACT [16], which handles MPCC complementarity via exact analytical projection. The contact-implicit IPOPT baseline instead uses Scholtes relaxation [27]. A trial is successful if its final goal error is below 0.01, and reported solve times include dynamics evaluation.

Effect of Lifted Multiple Shooting. Table II shows that lifted multiple shooting substantially improves the reliability of contact-rich trajectory optimization compared with conventional shooting formulations. With IPOPT, lifted multiple shooting achieves success rates of 98.5%, 95.5%, 100%, 99.5%, 94.5%, and 95.5% for box pushing, circle pushing, square knob manipulation, planar finger manipulation, sphere in-hand manipulation, and pen in-hand manipulation, respectively. In comparison, conventional multiple shooting achieves success rates of only 40.0%, 33.5%, 61.5%, 89.5%, 13.0%, and 20.0% on the same tasks. The improvement is also reflected in the final goal error, with lifted multiple shooting consistently obtaining substantially smaller errors.

The improved performance is consistent with the role of

the lifted contact impulses as explicit optimization variables. Rather than requiring the outer optimizer to modify the trajectory while contact impulses are determined implicitly through contact dynamics, the lifted formulation exposes the impulses directly and enforces their consistency through the equality constraint $\gamma_t = f_{\text{step1}}(x_t, u_t)$. This introduces additional decision variables and contact-resolution evaluations, but provides the optimizer with direct variables through which contact interactions can be adjusted. In particular, during dynamically infeasible intermediate iterates, the lifted impulse can directly modify the trajectory while the consistency constraint $\gamma_t = f_{\text{step1}}(x_t, u_t)$ is progressively restored, reducing the dependence of trajectory improvement on weak contact sensitivities near contact onset. The resulting formulation therefore trades a modest increase in computational cost for substantially improved optimization reliability.

Comparison with Contact-Implicit Optimization. Lifted multiple shooting achieves success rates comparable to the contact-implicit baselines. In particular, its performance is close to that of IMPACT, which enforces complementarity constraints in an MPCC using exact analytical projection. For box pushing, circle pushing, square knob manipulation, and planar finger manipulation, IMPACT achieves success rates of 96.0%, 100%, 95.5%, and 93.0%, respectively, compared with 98.5%, 95.5%, 100%, and 99.5% for IPOPT with lifted multiple shooting. Notably, lifted multiple shooting successfully transitions between contact modes induced by the nonsmooth geometry in both box pushing and square knob manipulation. This suggests that successful optimization does not require explicitly enumerating or encoding the individual faces of the contact geometry. Instead, the contact-resolution problem and the lifted impulse variables allow the optimizer to traverse these mode transitions through the underlying contact model. We do not evaluate IMPACT on the in-hand manipulation tasks because its implementation requires symbolic dynamics. The contact-implicit IPOPT formulation performs substantially worse, particularly on circle pushing and planar finger manipulation. It relaxes the complementarity constraints, whereas IMPACT handles them directly. These results suggest that lifted multiple shooting can achieve reliability comparable to a contact-implicit MPCC solver without introducing complementarity constraints into the outer optimization. Contact-implicit methods can nevertheless be faster: IMPACT avoids the repeated numerical contact-dynamics evaluations required by LMS. Thus, the reported runtimes reflect differences in both formulation and computational backend, not optimization cost alone.

IP-PAS Temporal Parallelization. We next compare IP-PAS with serial FATROP using the same lifted multiple shooting formulation. This comparison isolates the effect of replacing FATROP’s serial temporal recursion with the proposed parallel associative scan. IP-PAS achieves modestly lower solve times on the pushing and planar manipulation tasks, with a substantially larger reduction on the high-dimensional in-hand manipulation tasks. For example, on sphere in-hand manipulation, the mean solve time decreases from 12.6 ± 4.21 s with FATROP to 4.80 ± 2.29 s with IP-



Fig. 4: Hardware push box experiment.

PAS. The reduction in solve time does not come at the expense of solution quality. IP-PAS achieves success rates and goal errors comparable to FATROP across the evaluated tasks. For example, on sphere in-hand manipulation, IP-PAS achieves a 92.0% success rate and a mean goal error of 0.0242, compared with 90.5% and 0.0211 for FATROP. Similar behavior is observed for the pushing and planar manipulation tasks, where the differences in success rate and goal error remain small despite the reduced solve time. We additionally solve the longer-horizon in-hand sphere manipulation (Fig. 1a) and pen reorientation (Fig. 1b) tasks. These problems use 40- and 200-step horizons, respectively, and IP-PAS solves them in 2.8 and 7.3 seconds, respectively. Figure 3 further isolates the effect of temporal parallelization as the horizon grows on the square knob rotation task (Fig. 2c), where we scale the goal orientation such that the planner requires rotating the box by $\{5, 10, 20, 50, \text{and } 100\}$ revolutions and much longer planning horizons. While serial solve times grow substantially with horizon length, IP-PAS scales more gracefully to $T = 40000$, consistent with replacing the serial Riccati recursion by a logarithmic-depth parallel scan. This is well suited to the GPU, where stagewise contact dynamics and derivatives are already parallelized. The results show that PAS preserves the underlying interior-point optimization behavior, while larger gains on high-dimensional in-hand tasks indicate increasing benefits as stagewise computational cost grows.

Hardware Validation. We additionally validate the proposed approach on a physical box-pushing experiment. Figure 4 shows the robot executing a planned pushing trajectory on the hardware, successfully moving the box from its initial configuration toward the desired target. The experiment demonstrates that the trajectories generated by the proposed contact planning formulation can be transferred to a physical system.

VI. CONCLUSIONS

We presented a lifted multiple shooting transcription for trajectory optimization through contact together with a GPU-parallel interior-point solver based on a parallel associative scan. Across a range of contact planning and manipulation tasks, lifting contact impulses substantially improves optimization reliability over conventional single- and multiple-shooting transcriptions. The resulting formulation achieves success rates comparable to an MPCC-based contact-implicit solver while avoiding explicit complementarity constraints in the outer trajectory optimization problem. The proposed IP-PAS solver further preserves the success rate and solution quality of serial FATROP while reducing solve time, with the largest benefit observed for high-dimensional in-hand manipulation. Together, these results demonstrate the feasibility of explicitly exposing contact impulses and exploiting the temporal structure for trajectory optimization through contact.

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